Ilia Karmanov

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UK and EU citizen (no visa sponsorship needed)

Overview

Senior Staff Research Scientist at NVIDIA ADLR, working on post-training and evaluation of vision-language models for multimodal reasoning. First author of Eclair, a document-understanding model adopted across NVIDIA's pipelines for pre-training data extraction and VLM distillation, and shipped as Nemotron Parse. Earlier, published at NeurIPS, ICCV, and BMVC while at Qualcomm AI Research. Trained in economics at LSE, with a focus on optimal control and causal inference, before moving into machine learning in 2016.

Experience

NVIDIA ADLR, Zurich — Mar 2025 – present
Senior Staff Research Scientist

- Post-train and evaluate NVIDIA's vision-language models, Nemotron Nano V2 VL and Nemotron 3 Nano Omni, built on pre-trained LLM and vision-encoder backbones
- Run multi-node distributed post-training in PyTorch (100+ GPUs, Megatron/NeMo): GRPO and on-policy distillation for multimodal and long-context reasoning (e.g. MMLongBench), with reward and objective design for visual grounding

NVIDIA ADLR, Zurich — Sep 2022 – Feb 2025
Staff Research Scientist

- First author of Eclair, a document-understanding model that extracts formatted text, tables, formulas, and reading order. Shipped open-source as Nemotron Parse and adopted across NVIDIA's pipelines for pre-training data (50B+ tokens) and VLM distillation; it underpinned NVIDIA's Llama Nemotron Nano VL, which ranked first on the OCRBench v2 benchmark
- Led the model's design: an asymmetric architecture (larger vision encoder, lightweight decoder), no positional embeddings in the decoder, multi-token prediction to reduce decoding steps, and a single prompt-controlled output format spanning text, layout, and semantic classes
- Contributed to the token-compression design for dense OCR in Eagle 2 (NeurIPS 2025). Mentored two interns (both later joined full-time)

Qualcomm AI Research, Amsterdam — 2021 – 2022
Staff Research Scientist

- Developed hardware-efficient architectures for 4K video-to-video processing (denoising on device)
- Efficient Mixture-of-Experts architecture for conditional computation (BMVC 2022)

Qualcomm AI Research, Amsterdam — 2020 – 2021
Senior Research Scientist

- Created the first weakly-supervised approach for 3D person-localisation on video and WiFi data. Published at NeurIPS 2021 (with Max Welling) and IEEE GLOBECOM 2021 (first author)
- Video recognition via motion-augmented self-training (ICCV 2021, with Cees Snoek)

Microsoft, London — 2018 – 2020
Senior Applied Scientist

- Applied computer vision across industry problems: self- and fully-supervised segmentation (medical and seismic imaging), object detection and counting, and real-time video action recognition
- Synthetic-data pipelines using inductive biases (low-frequency and shape signals) to improve generalisation

Microsoft, London — 2016 – 2018
Applied Scientist

- Collaborated with MSR on Z3 theorem-prover for plane-gate allocations (Dubai Airport)
- Created DeepLearningFrameworks: benchmarking deep learning frameworks across single-node and multi-node on Azure. 1,700+ GitHub stars, contributions from framework creators

Charles River Associates; IGC (Oxford); LSE 2013 – 2016

Research Economist / Consultant

- Econometrics for antitrust litigation (Charles River Associates); firm-capability and trade research at the Oxford centre directed by Prof. Paul Collier (IGC); and causal-inference (diff-in-diff) work as RA to Prof. Frank Cowell (LSE), leading to a published paper (Costa-Font & Cowell, 2015)

Publications

Authors incl. **I Karmanov**. NVIDIA Nemotron 3 Nano Omni: Efficient and Open Multimodal Intelligence. arXiv 2026

I Karmanov, A Deshmukh, L Voegtle, P Fischer, K Chumachenko, T Roman, J Seppanen, J Parmar, J Jennings, A Tao, K Sapra. Eclair: Extracting Content and Layout with Integrated Reading Order for Documents. arXiv 2025

K Chumachenko, A Deshmukh, J Seppanen, **I Karmanov**, C Chen, L Voegtle, P Fischer, et al. Nemotron Parse 1.1. arXiv 2025

140+ authors incl. **I Karmanov**. NVIDIA Nemotron Nano V2 VL. arXiv 2025

199 authors incl. **I Karmanov**. Nemotron-H: A Family of Accurate and Efficient Hybrid Mamba-Transformer Models. arXiv 2025

Z Li, G Chen, S Liu, ... **I Karmanov**, L Voegtle, P Fischer, ... Z Yu. Eagle 2: Building Post-Training Data Strategies from Scratch for Frontier Vision-Language Models. NeurIPS 2025

A Royer, **I Karmanov**, A Skliar, B Ehteshami Bejnordi, T Blankevoort. Revisiting Single-gated Mixtures of Experts. BMVC 2022

FG Zanjani, **I Karmanov**, H Ackermann, D Dijkman, S Merlin, M Welling, F Porikli. Modality-Agnostic Topology Aware Localization. NeurIPS 2021

I Karmanov, FG Zanjani, S Merlin, I Kadampot, D Dijkman. WiCluster: Passive Indoor 2D/3D Positioning using WiFi without Precise Labels. IEEE GLOBECOM 2021

K Gavrilyuk, M Jain, **I Karmanov**, CGM Snoek. Motion-Augmented Self-Training for Video Recognition at Smaller Scale. ICCV 2021

Y Ren, J Lu, A Beletchi, Y Huang, **I Karmanov**, et al. Hand Gesture Recognition using 802.11ad mmWave. IEEE WCNC 2021

Book: M Salvaris, D Dean, WH Tok, **I Karmanov**. Deep Learning with Azure. Apress, 2018

J Costa-Font, F Cowell (with research contributions from **I Karmanov**). European Identity and Redistributive Preferences. CESifo Working Paper, 2015

Patents

12 US patent applications filed at Qualcomm (several granted), across RF positioning, gesture recognition, and efficient architectures.

Service

Reviewer for NeurIPS, ICLR, and ICML.

Education

London School of Economics 2012 – 2013

MSc Economics

- Thesis: Optimal Control Analysis for Tax Avoidance. Modelling a firm's optimisation problem in an infinite horizon setting, introducing reputation as an accumulated asset, solved with Hamiltonian in an optimal-control setting. Advisor: Prof. Frank Cowell

London School of Economics 2009 – 2012

BSc Economics, First Class Honours